CKD Precision Medicine: Project Exploration Guide

The Core Problem (All Angles Address This)

Clinical Reality:

- 37 million Americans have CKD
- 500K on dialysis (expensive, low quality of life)
- Progression is UNPREDICTABLE
- Some stable 20+ years, some reach dialysis in 2-3 years
- Doctors have no framework to distinguish

Current Approach (Broken):

```
Patient: "I have CKD stage 3"
Doctor: "Keep your BP controlled, see you in 6 months"
Patient: "Will I need dialysis?"
Doctor: "I don't know. Maybe."
```

Why This Sucks:

- Slow progressors get over-treated (unnecessary aggressiveness)
- Fast progressors get under-treated (miss intervention window)
- Healthcare is reactive, not predictive

Your Role: Make it predictive. Use ML to tell doctors: "THIS patient will progress fast, needs aggressive treatment NOW" or "THIS patient is stable, monitor normally"

Project Idea #1: Basic Progression Risk Prediction

What You're Building

Train ML model to predict: Fast vs Slow progression

Input: Baseline patient data

- Age, sex, race
- eGFR (kidney function measure)
- Proteinuria (protein in urine)

- Blood pressure
- Comorbidities (diabetes, hypertension)
- Medications
- Labs (albumin, glucose, sodium)

Output: Risk prediction

- HIGH / MODERATE / LOW
- Probability: "78% will progress fast"
- Top 3 drivers: "Proteinuria, age, and diabetes"

Data Source

Option A: MIMIC (Easiest Start)

```
- 50K+ CKD patients
- Serial creatinine measurements over time
- BP, medications, comorbidities
- Outcomes: Did they start dialysis?
- Cost: FREE (PhysioNet)
- Access: 3-4 days (credential + CITI training)
- Problem: ICU-focused, may not have long follow-up
- Time to data: FASTEST (1 week)
```

Option B: Published CKD Cohorts (Better Quality)

```
SPRINT Trial:
- 9,000 CKD patients
- 5-year kidney outcomes
- Treatment arms (helps validate)
- Access: biolincc.nhlbi.nih.gov
- Problem: Data use agreement required (slower)
- Time to data: 2-4 weeks

ARIC Study:
- 15,000 participants
- 30-year follow-up
- Serial kidney function
- Access: dbGaP.ncbi.nlm.nih.gov
- Problem: Complex database
- Time to data: 2-4 weeks
```

Option C: Synthetic Data (Backup)

```
- Build realistic CKD cohort (10K patients)
- Based on published epidemiology
- Use: Python faker + numpy
- Advantage: Full control, instant access
- Disadvantage: Not real patient data
- Use if: Can't access real data quickly
```

Validation Approach

Step 1: Beat Current Standard

```
KDIGO Risk Calculator: AUC 0.68
Your model target: AUC > 0.78
Validation method: 70% train, 30% test (held-out patients)
Judge sees: "I'm better than what doctors currently use"
```

Step 2: Sensitivity/Specificity

```
Can you catch 80%+ of fast progressors?
Can you avoid false alarms? (specificity > 70%)
Judge sees: "This is clinically useful"
```

Step 3: External Validation

```
If using MIMIC: Test on SPRINT cohort
If using SPRINT: Test on ARIC
Validates: Does it work across populations?
Judge sees: "This generalizes, not overfit"
```

Step 4: Clinical Utility

```
High-risk patients left untreated: X% reach dialysis
High-risk patients treated aggressively: Y% reach dialysis
Show: Difference = lives saved
Judge sees: "This actually prevents disease"
```

Timeline

```
Week 1:    Data access + exploration (4 days)
```

```
Week 2-3:  Feature engineering + EDA (5 days)
Week 4-5:  Model training + hyperparameter tuning (10 days)
Week 6:    Validation on test set (3 days)
Week 7:    External validation (3 days)
Week 8:    SHAP explainability (3 days)
Week 9:    Write paper + clinical utility analysis (5 days)
Week 10:   ISEF poster/presentation (3 days)
Week 11:   Practice + refinement (3 days)
Week 12:   Final submission + buffer (3 days)
```

How to Explain to Judges (60 seconds)

```
"Half of CKD patients progress slowly.
Half progress fast and need dialysis.
Doctors don't know which is which.

I built an AI model that predicts which patients will progress fast
using their baseline kidney function and medical history.

Result: Doctors can identify high-risk patients upfront
and treat them aggressively, preventing dialysis in thousands
of patients per year."
```

Difficulty & Novelty

```
Difficulty: MEDIUM (straightforward ML)
Novelty: MEDIUM (risk prediction exists, but not well for CKD)
Feasibility: HIGH (data accessible, well-defined problem)
Judge Appeal: HIGH (clear problem, clear solution)
```

Pros & Cons

Pros:

- Clear problem statement
- Objective success metrics (AUC, sensitivity, specificity)
- Real clinical impact
- Straightforward ML (judges understand)
- Lots of precedent (other ML risk scores)

Cons:

- Not super novel (risk prediction is standard)
- Judges may have seen similar projects
- Relatively simple ML (not showcase technical sophistication)
- May feel "just applying techniques to new domain"

Project Idea #2: Personalized Treatment Response Prediction

What You're Building

Don't just predict progression rate.
Predict: Which treatment will work BEST for THIS patient?

Input: Baseline patient data + treatment info

- Same as above PLUS
- What treatment are they considering? (BP control intensity, SGLT2i, GLP-1, etc)

Output: Personalized recommendation

- "For YOU: SGLT2i will slow progression 40%"
- "For YOU: Aggressive BP control will slow progression 20%"
- "For YOU: Standard therapy will slow progression 5%"

The Science

Reality: CKD treatment response varies wildly

- Some patients respond dramatically to SGLT2i
- Others barely respond
- Some benefit from aggressive BP control
- Others don't

Why: Underlying kidney disease varies

- Diabetic kidney disease (SGLT2i works great)
- Hypertensive kidney disease (BP control critical)
- IgA nephropathy (immunosuppression needed)
- Polycystic kidney disease (tolvaptan works)

Data Source

Option A: MIMIC + Treatment Outcomes

```
Extract:
- CKD patients
- What treatment they received
- How did they respond? (eGFR change)
- Correlate: baseline features + treatment + response

Challenge: MIMIC has lots of CKD, limited "planned" treatments
Mostly acute care, not outpatient management
```

Option B: Published Trial Data

```
SPRINT Trial (BP control in CKD):
- Some randomized to intensive BP (SBP <120)
- Some randomized to standard (SBP <140)
- Outcomes: kidney disease progression

EMPA-KIDNEY Trial (SGLT2i in CKD):
- Randomized to empagliflozin vs placebo
- Outcomes: eGFR decline

Your approach:
- Combine data from multiple trials
- Learn: who responds to SGLT2i vs BP control vs other
```

Option C: Observational Cohort + Treatment Records

```
Build cohort of CKD patients
Record: baseline features + treatments they received + outcomes
Learn: treatment-outcome relationships
```

Validation Approach

Step 1: Predict Treatment Response Accurately

```
For patients receiving SGLT2i:
- Predict: eGFR decline with drug
- Actual: measure eGFR change
- Validate: prediction matches reality (R² > 0.6)
```

Step 2: Predict Who Benefits Most

```
Patients with diabetes + CKD:
- Some respond dramatically to SGLT2i
- Some don't respond much
- Can you predict which?
Validation: compare predicted responders to actual responders
```

Step 3: Show Matching Works Better

```
Scenario A: Give all CKD patients SGLT2i
Result: 50% have significant benefit

Scenario B: Use your model to predict responders, only treat those
Result: 80% have significant benefit (same drug, better patient selection)
```

Timeline

```
Week 1-2:  Data wrangling (combine multiple datasets)
Week 3-4:  Feature engineering + phenotype discovery
Week 5-6:  Build model: predict treatment response
Week 7:    Validate on trial outcomes
Week 8:    Compare: personalized vs standard assignment
Week 9:    Analysis + paper
Week 10-12: Presentation
```

How to Explain to Judges (60 seconds)

```
"Current CKD treatment is one-size-fits-all.
All patients get same drugs, some respond, some don't.

I discovered that different CKD patients have different underlying
kidney disease types. Each responds best to different treatments.

I built AI that:
1. Identifies which type of kidney disease YOU have
```

```
2. Recommends the treatment most likely to work for YOU

Result: Instead of trial-and-error over months,
doctors start with the right treatment first."
```

Difficulty & Novelty

```
Difficulty: MEDIUM-HIGH (more complex data, heterogeneous treatment effects)
Novelty: MEDIUM-HIGH (treatment matching in CKD is less studied)
Feasibility: MEDIUM (requires combining multiple data sources)
Judge Appeal: HIGH (personalization is appealing)
```

Pros & Cons

Pros:

- More novel than simple prediction
- Clear personalization angle (judges love precision medicine)
- Real clinical impact (right drug for right patient)
- Shows advanced understanding (heterogeneous treatment effects)

Cons:

- More complex data wrangling
- Harder to validate (need outcome data post-treatment)
- May require longer follow-up data
- Risk prediction vs treatment matching is conceptually harder

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Project Idea #3: Mechanistic CKD Phenotyping

What You're Building

Instead of: "predict progression"
Ask: "what's the underlying cause of THIS patient's kidney disease?"

Biological Phenotypes:

1. Diabetic kidney disease (high glucose damages glomeruli)
2. Hypertensive kidney disease (high BP damages glomeruli)
3. Glomerulonephritis (immune attack on kidney)

4. Cystic kidney disease (structural abnormality)
5. Tubular disease (proximal tubule dysfunction)
6. Vascular disease (small vessel disease)
7. Other (medication-induced, infection, etc)

Your ML Task: Input: Clinical features, labs, imaging Output: What phenotype? → Dictates treatment strategy

Data Source

Option A: Published Pathology + Clinical Correlation Studies

```
Search: "kidney biopsy" + "clinical correlation" + machine learning
Many papers have biopsy findings + baseline clinical data
Extract: histology findings + baseline features
Learn: which clinical features predict biopsy findings?

Advantage: Objective ground truth (biopsy is diagnostic)
Disadvantage: Only patients who had biopsies (selection bias)
```

Option B: Imaging + Clinical Features

```
Use: Kidney ultrasound/CT findings + labs
- Cystic disease: obvious on imaging
- Diabetic: correlated with diabetes + proteinuria
- Hypertensive: correlated with hypertension
- Inflammatory: correlated with inflammatory markers + hematuria
Learn: predict phenotype without biopsy
```

Option C: Lab Signatures (No Biopsy)

```
Different phenotypes have different lab profiles:
- Diabetic: albuminuria, blood glucose high
- IgA: hematuria present
- Lupus: ANA+, complement low
- Vasculitis: ANCA+, proteinuria, hematuria

Your model: Lab pattern → CKD phenotype
Advantage: Works without biopsy (cheaper, faster)
```

Validation Approach

Step 1: Predict Biopsy Findings

```
For patients who had kidney biopsies:
- Predict: what will biopsy show?
- Actual: pathology report
- Validate: accuracy > 70%
```

Step 2: Predict Underlying Cause

```
For patients diagnosed with specific CKD type:
- Predict: from baseline labs
- Actual: confirmed diagnosis
- Show: better than current approach
```

Step 3: Show Clinical Usefulness

```
Traditional: Doctor suspects diabetic kidney disease, maybe refers for biopsy
Your model: "95% confident this is diabetic disease, no biopsy needed"

Result: Fewer unnecessary biopsies, faster diagnosis, immediate treatment
```

Timeline

```
Week 1-2:  Literature search + data compilation
Week 3-4:  Feature engineering (which labs predict phenotype?)
Week 5-6:  Build classifier (phenotype prediction)
Week 7:    Validate on biopsy cohort
Week 8:    Show clinical utility (diagnostic accuracy)
Week 9-10: Paper + presentation
```

How to Explain to Judges (60 seconds)

```
"Kidney disease has many causes: diabetes, high blood pressure, infections, etc.
Each cause needs different treatment.

But many patients don't get a kidney biopsy to diagnose the cause.
So doctors guess and treat empirically.

I built AI that diagnoses the CAUSE of kidney disease from blood/urine tests,
without needing a biopsy.
```

```
Result: Doctors can start treatment for the RIGHT cause immediately,
instead of guessing and adjusting later."
```

Difficulty & Novelty

```
Difficulty: MEDIUM-HIGH (requires understanding kidney pathology)
Novelty: MEDIUM-HIGH (kidney disease phenotyping is emerging area)
Feasibility: MEDIUM (good datasets exist, conceptually complex)
Judge Appeal: MEDIUM-HIGH (mechanistic understanding + practical impact)
```

Pros & Cons

Pros:

- More mechanistic (not just correlational)
- Shows deep understanding of disease biology
- Very practical (guides diagnosis)
- Novel in ML literature (rare to see this)

Cons:

- Requires understanding kidney pathology (steep learning curve)
- Data may be harder to find (biopsy cohorts are smaller)
- Conceptually more complex to validate
- Judges may not appreciate the nuance

Project Idea #4: Drug Toxicity Risk Prediction for CKD

What You're Building

CKD patients on multiple medications.
Some drugs are toxic in kidney disease (need dose adjustment).

Problem:

- Patient takes 5 medications
- Kidneys failing (low eGFR)
- Accumulate toxic levels of drugs
- Get adverse effects

- Doctor doesn't realize it's the drug

Your solution: Predict: which CKD patients at risk for drug toxicity?

Input: Patient eGFR + medications **Output:** Risk prediction for each drug

- "Metformin: AVOID (high lactic acidosis risk)"
- "Gentamicin: CAUTION (needs dose adjustment to 10mg Q48H, not 5mg Q24H)"
- "Lisinopril: SAFE (but monitor K+)"

Data Source

Option A: MIMIC + Adverse Events

```
Query: CKD patients taking medications + lab derangements
- Medication X + creatinine Y → hyperkalemia?
- Medication A + eGFR B → acute injury?
Learn: which drugs problematic at which eGFR levels

Challenge: Hard to distinguish "drug caused it" vs "disease caused it"
```

Option B: FDA Adverse Event Database (FAERS)

```
15M+ adverse event reports
Filter: CKD patients taking medications
Extract: medication + adverse outcome
Learn: toxicity risk by eGFR

Advantage: Large, real-world adverse events
Disadvantage: Confounded (hard to know causation)
```

Option C: Clinical Knowledge + Data

```
Start with: Published drug dosing guidelines
- Which drugs contraindicated in CKD?
- Which need dose adjustment?
- What eGFR thresholds?

Validate with: MIMIC or FAERS data
```

Validation Approach

Step 1: Predict Known Toxicities

```
For drugs with well-known CKD toxicity (e.g., aminoglycosides):
- Predict: toxicity risk at different eGFR levels
- Actual: published guidelines
- Show: your model matches guidelines
```

Step 2: Find Unknown Toxicities

```
Mine FAERS: find drug-toxicity pairs not in guidelines
Example: "Clarithromycin in CKD associated with hyperkalemia"
Validate: Is this real? (Check literature, other databases)
```

Step 3: Show Clinical Impact

```
High-risk patients identified by model:
- If doctors dose-adjust: adverse events ↓
- If doctors ignore: adverse events stay high
```

Timeline

```
Week 1-2:  Compile drug dosing guidelines + FAERS data
Week 3-4:  Build toxicity prediction model
Week 5:    Validate against known guidelines
Week 6:    Search for novel toxicities
Week 7:    Validate novel findings
Week 8-10: Paper + presentation
```

How to Explain to Judges (60 seconds)

```
"CKD patients take multiple drugs.
But kidneys can't clear drugs as fast.
Drugs build up to toxic levels.

Doctors sometimes forget to adjust doses.
Patient gets toxicity (hyperkalemia, lactic acidosis, etc).

I built AI that predicts which CKD patients are at risk
for toxicity from their medications.

Result: Doctors get alerts to dose-adjust or switch drugs,
preventing hospitalizations."
```

Difficulty & Novelty

```
Difficulty: MEDIUM (pharmacokinetics + toxicology)
Novelty: MEDIUM-HIGH (rarely applied systematically)
Feasibility: HIGH (data accessible, clear problem)
Judge Appeal: MEDIUM (specific, practical, not as sexy as others)
```

Pros & Cons

Pros:

- Very practical (directly actionable)
- Clear success metric (prevent adverse events)
- Data very accessible (FAERS, guidelines)
- Relatively quick to build

Cons:

- Narrower scope (just toxicity, not progression)
- Less flashy (less "sexy" than other ideas)
- Validation harder (causation vs correlation in adverse events)
- Judges may think it's "just applying rules"

Project Idea #5: Predict ESRD Timing (When Will Dialysis Be Needed?)

What You're Building

Not just "will you progress fast" but "WHEN will you need dialysis?"

Input: Baseline patient data + longitudinal eGFR trajectory **Output:** Time prediction

- "This patient will reach ESRD in 3 years ± 1 year (95% CI)"
- "Versus median 7 years for similar patients"

Why This Is Hard (and therefore Novel)

Standard regression: Predict time-to-event **Challenge:** Censoring (some patients don't reach ESRD in study period) **Requires:** Survival analysis (Kaplan-Meier, Cox regression, random survival forests)

Data Source

Same as #1 but with longer follow-up

```
MIMIC: May not have 5+ year follow-up
SPRINT: 5-year outcomes, perfect
ARIC: 30-year follow-up, ideal

Requirement: Know when (if) patient reached dialysis
```

Validation Approach

Step 1: Calibration

```
Predict: "75% will reach ESRD in 3 years"
Actual: Did 75% actually reach ESRD in 3 years?
Validation: Calibration plot (predicted vs actual)
```

Step 2: Discrimination

```
Can you separate fast progressors from slow?
C-index > 0.75 (like AUC but for survival)
```

Step 3: Clinical Utility

```
"With timing prediction, doctors can plan:
- Fistula placement (needs 3-6 months maturation)
- Kidney transplant workup
- Lifestyle changes"
```

Timeline

```
Week 1-2:  Data + survival analysis concepts
Week 3-4:  Build survival model (random survival forest, Cox)
Week 5:    Validate calibration
Week 6:    Validate discrimination
Week 7-10: Paper + presentation
```

How to Explain to Judges (60 seconds)

```
"Doctors tell CKD patients: 'You might need dialysis sometime.'
Patient: 'When?'
Doctor: 'I don't know.'
```

```
I built AI that predicts WHEN you'll need dialysis,
based on your current kidney function and other factors.

Result: Patients can plan ahead (surgery, jobs, transplants)
and doctors can time interventions optimally."
```

Difficulty & Novelty

```
Difficulty: MEDIUM-HIGH (survival analysis is more complex)
Novelty: HIGH (rarely done for CKD)
Feasibility: MEDIUM (requires long follow-up data)
Judge Appeal: HIGH (specific, actionable, novel)
```

Pros & Cons

Pros:

- Novel (rare in ML literature)
- Very specific/actionable (when, not just if)
- Shows advanced statistical techniques
- Judges impressed by survival analysis

Cons:

- Requires longer follow-up data (harder to get)
- More complex statistical concepts (harder to explain)
- Smaller effect sizes (harder to validate)
- Judges may not understand survival analysis

Comparison Table

Project	Data Access	Difficulty	Novelty	Judge Appeal	Timeline	Best For
#1: Progression Risk	EASIEST	MEDIUM	MEDIUM	HIGH	10 weeks	Safe choice, clear problem

Project	Data Access	Difficulty	Novelty	Judge Appeal	Timeline	Best For
#2: Treatment Response	MEDIUM	MEDIUM-HIGH	MEDIUM-HIGH	HIGH	11 weeks	Precision medicine angle
#3: Mechanistic Phenotyping	MEDIUM	MEDIUM-HIGH	MEDIUM-HIGH	MEDIUM	10 weeks	Disease understanding
#4: Drug Toxicity	EASY	MEDIUM	MEDIUM-HIGH	MEDIUM	9 weeks	Practical + actionable
#5: ESRD Timing	MEDIUM	MEDIUM-HIGH	HIGH	HIGH	10 weeks	Advanced statistics

How to Choose

Choose #1 (Progression Risk) if:

- You want safest choice
- You want clearest validation
- You want quickest implementation
- You're uncomfortable with statistics

Choose #2 (Treatment Response) if:

- You love precision medicine
- You want "personalization" angle
- You're comfortable with advanced concepts
- You want judges impressed

Choose #3 (Phenotyping) if:

- You want to show disease understanding
- You like mechanistic explanations
- You're willing to learn kidney pathology
- You want most "scientific" project

Choose #4 (Drug Toxicity) if:

- You want narrow, focused scope
- You like practical problems
- You want quick validation
- You prefer "solvable" over "flashy"

Choose #5 (ESRD Timing) if:

- You're strongest at stats/ML
- You want most novel project
- You're willing to explain complex concepts
- You want to impress ISEF judges most

What I Recommend

For ISEF winning + Real impact + Doable:

#1 → #2 pathway

Start with #1 (Progression Risk Prediction)

- Easiest data access
- Clearest validation
- Solid ISEF project
- 10 weeks

THEN (if you have time/interest):

- Extend to #2 (Treatment Response)
- Add personalization layer
- "Better" for judges
- +2 weeks

Result: Strong project in 12 weeks, or exceptional project in 14 weeks

---

Next Steps

1. **Pick your angle** (which resonates most?)
2. **Start data access** (MIMIC or download published cohort)
3. **Week 1-2 task:** Explore data, write feature engineering code
4. **Check back:** Show me what you find in the data

Which one are you most excited about?

Or want me to deep-dive one of these further?